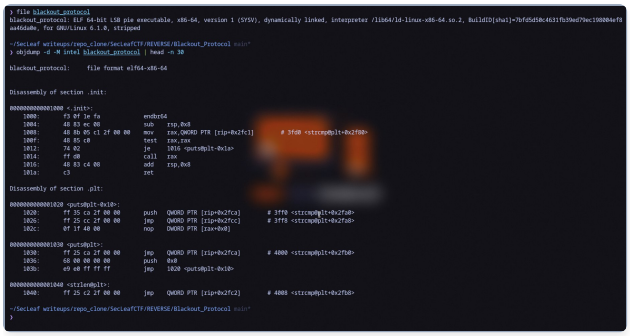
Blackout Protocol

Reverse Engineering



I started by checking the challenge archive containing stripped 64-bit ELF binary `blackout_protocol` and an intercepted network trace.

Decompiling in Ghidra and inspecting GDB section headers revealed custom packet structures and a command dispatch table handling state synchronization across virtual substations.

The protocol used dynamic XOR masks derived from consecutive sequence counters combined with CRC16 frame validation. I reversed the substation authentication handshake and payload decryption routine to extract the master override command.

Finally, I tested the recovered flag against the executable / service:

`python3 Blackout_Protocol.py`

Substation handshake validated. Frame decoded.

Flag Captured

`SecLeaf{b14ck0ut_pr0t0c01_0v3rr1d3_5ucc355}`

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